

SPACE X

Predicting Falcon 9 First-Stage Landings

Applied Data Science Capstone • Technical Report

A reproducible pipeline from API and web data to interactive analytics and classification.

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<https://github.com/ijontichyai/IBM-Data-Science-Capstone>

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Executive summary

Can launch history reveal when a Falcon 9 first stage is likely to land?

90

launch records analyzed

76.9%

KSC LC-39A success rate

84.8%

best cross-validation score

83.3%

verified holdout accuracy

What we did

- Collected launch data via API and web scraping
- Cleaned, encoded, queried, mapped, and visualized the data
- Tuned Logistic Regression, SVM, Decision Tree, and KNN

What we learned

- Launch site, payload, orbit, and booster context matter
- SVM led validation; SVM and Logistic Regression tied on test data
- False positives—not missed landings—were the main error mode

Business context and analytical problem

WHY IT MATTERS

First-stage reuse can reduce launch cost and improve mission economics.

A reliable landing prediction helps planners estimate reuse potential before launch.

Research question

Given launch conditions and historical outcomes, will the first stage land successfully?

INPUTS

Payload • Orbit • Site • Booster • Flight history

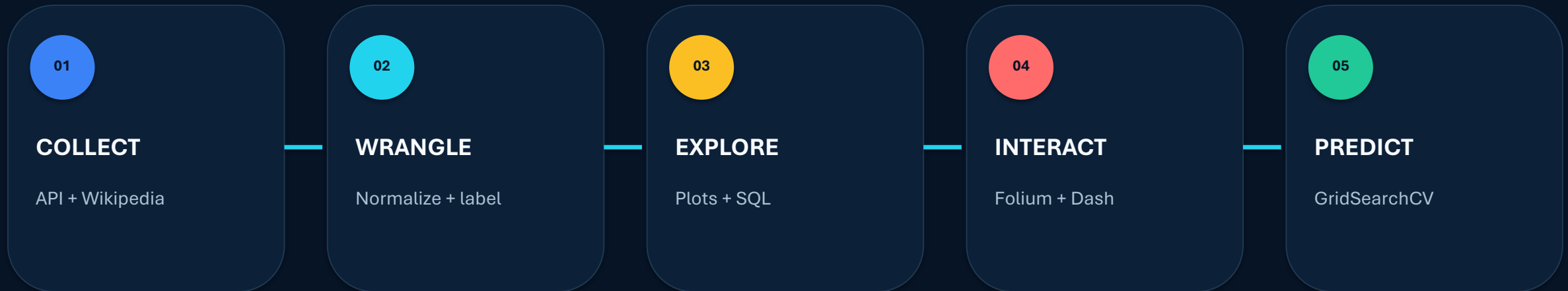
TARGET

Class = 1 landed | Class = 0 did not land

OUTPUT

Transparent evidence for operational decision-making

From raw launch records to decision evidence



Design principle

Every result is traceable to source data, transformations, and a reproducible notebook or Python file.

Data collection — SpaceX REST API

GET`/v4/launches/past`**LOOK UP**`rockets · payloads ·
launchpads · cores`**NORMALIZE**`JSON → pandas
DataFrame`**FILTER**`Falcon 9 launch records`**EXPORT**`structured CSV`

Core fields retained

- Flight number and date
- Booster / core configuration
- Payload mass and orbit
- Launch site and landing outcome

Quality controls

- Resolve related IDs before analysis
- Remove empty or irrelevant records
- Preserve a repeatable extraction path
- Store the final analytical table

Data collection — web scraping

- 1 Request Wikipedia launch-history page
- 2 Parse HTML tables with BeautifulSoup
- 3 Extract rows and normalize cell text
- 4 Build a structured launch-record DataFrame
- 5 Validate columns and export for analysis

Fields extracted

Flight No.

Launch site

Payload / orbit

Customer

Mission outcome

Booster landing

Data wrangling and feature engineering



CLEAN

Handle missing payloads, normalize text, remove duplicates



LABEL

Convert landing outcome to binary class 0 / 1



ENCODE

One-hot encode orbit, site, landing pad, and serial



SCALE

Standardize numeric features before distance-based models

Result: one analysis-ready feature matrix X and one target vector Y .

Exploratory analysis — questions before models

SITE

Where do launches succeed most often?

PAYLOAD

How does mass relate to outcome?

ORBIT

Which mission profiles are most successful?

BOOSTER

Which configurations perform best?

TIME

Does success improve as experience grows?

RISK

Where do failures cluster?

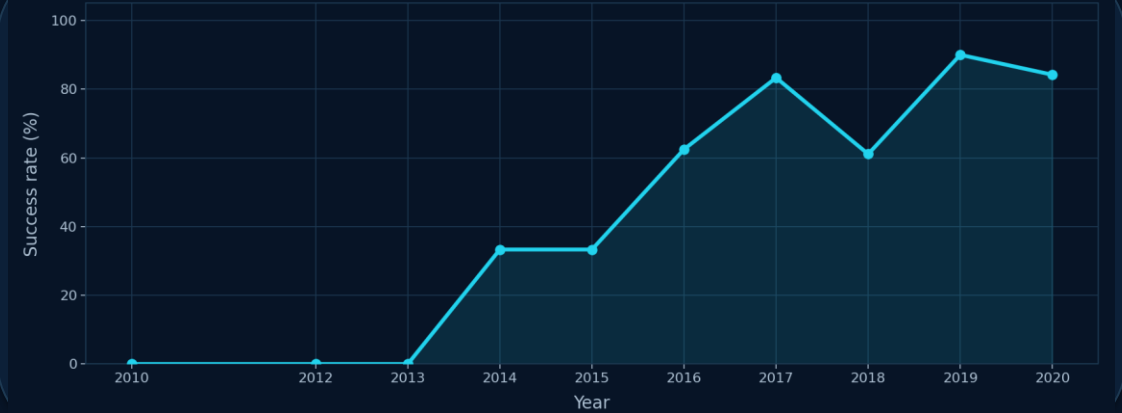
Bar charts compare site and booster performance; scatter plots examine payload vs. outcome; line charts show yearly success trends; SQL verifies exact aggregations.

EDA results — site, payload, and experience

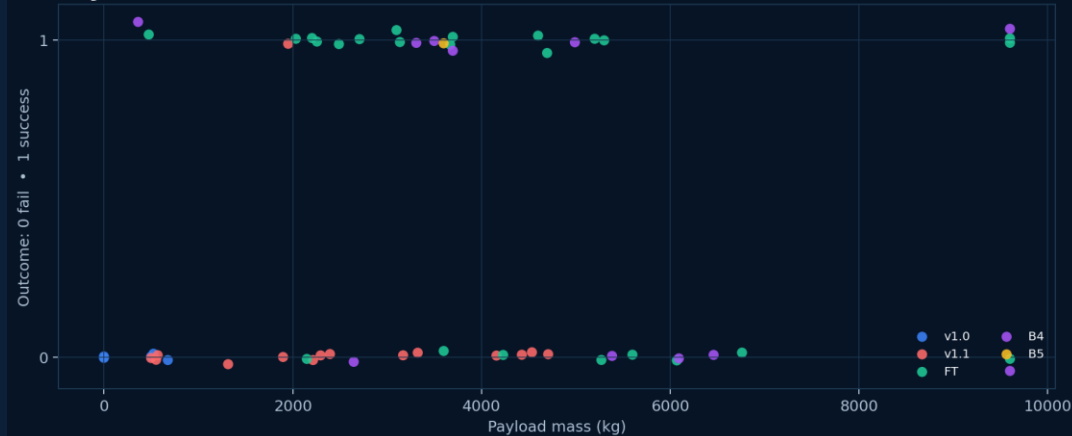
Successful launches by site



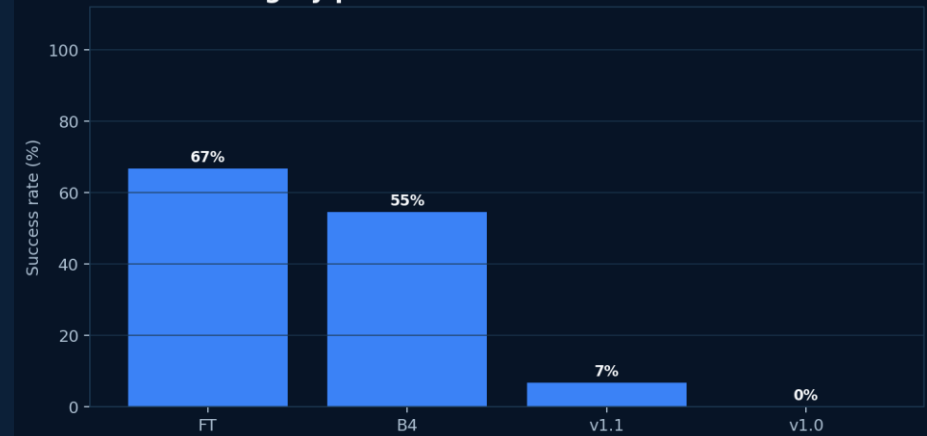
Launch success rate by year



Payload versus launch outcome



Booster category performance



EDA with SQL — reproducible analytical questions

- | | | |
|----|----------------------|--|
| 01 | SITE COVERAGE | DISTINCT launch sites and CCA-prefixed records |
| 02 | PAYLOAD | SUM and AVG payload mass by customer / booster |
| 03 | SUCCESS | COUNT outcomes and drone-ship landings |
| 04 | RANKING | GROUP BY booster and ORDER BY success count |
| 05 | TIME | Filter launch dates and compare historical periods |

Pattern: **SELECT** → **WHERE** → **GROUP BY** → aggregate → **ORDER BY**

SQL results — exact summaries support the story

QUESTION	RESULT	USE
Launch sites	Four sites represented in the analytical dashboard	Coverage
Payloads	Observed payload range extends to about 9,600 kg	Distribution
Success rates	KSC LC-39A: 76.9% • CCAFS LC-40: 73.1%	Performance
Ranking	KSC LC-39A contributes the largest share of successes: 41.7%	Ranking
Time analysis	Later launch sequences show stronger and more stable outcomes	Trend

Interactive geospatial analytics with Folium

MAP LAYERS

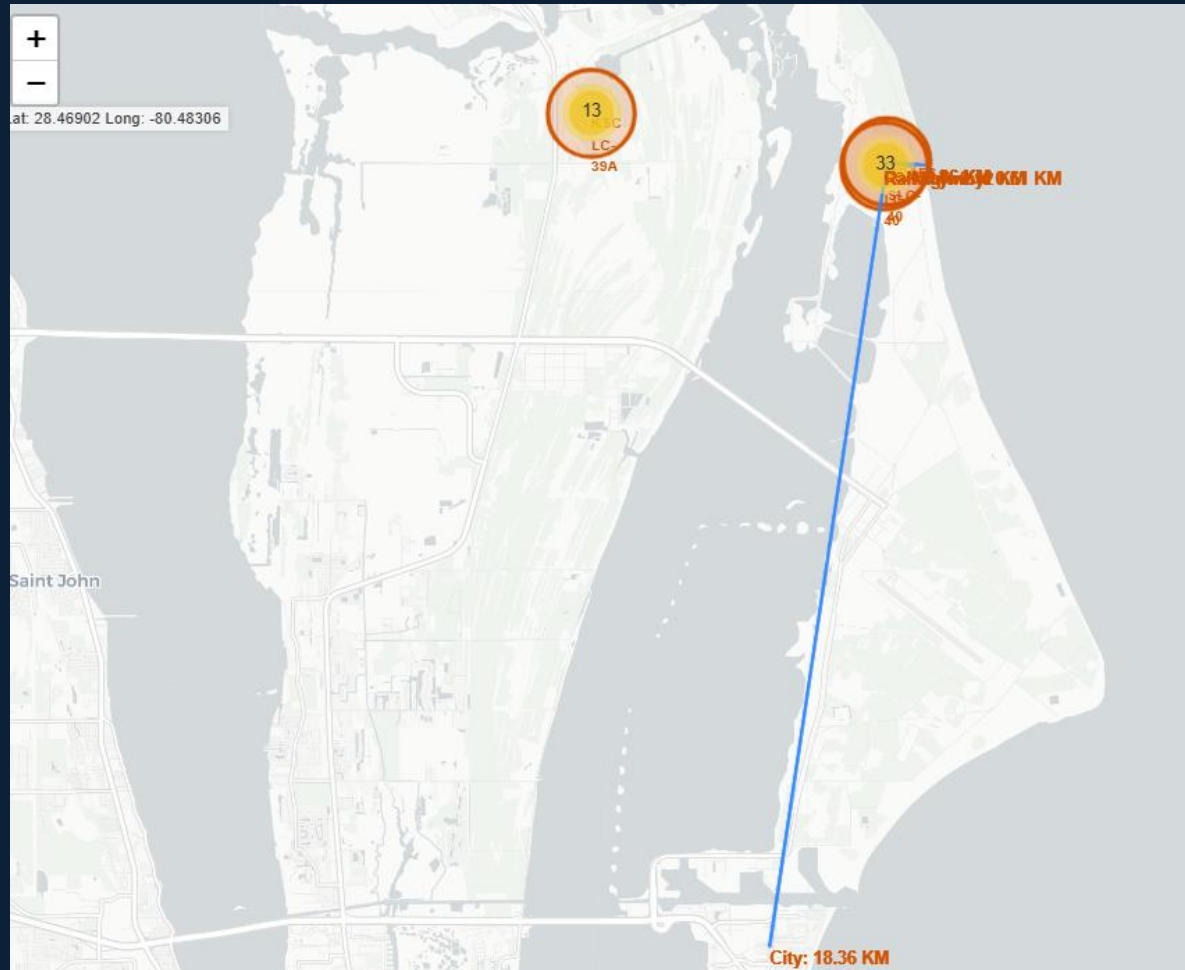
- 1 Launch sites** Circle markers + labels
- 2 Launch records** Green success / red failure markers
- 3 Clusters** MarkerCluster for overlapping launches
- 4 Proximity** MousePosition + distance lines

PROXIMITY TESTS

-  **Coastline**
-  **Road network**
-  **Railway**
-  **City / population center**

Goal: explain why launch facilities occupy their specific geography.

Folium results — coastal sites and controlled access



Key geographic findings

- Railway: 0.12 km
- Highway: 0.61 km
- Coastline: 0.86 km
- City: 18.36 km

Plotly Dash — turning analysis into an interface

INPUTS

Launch-site dropdown

Payload range slider

USER ACTION

Select a site and
constrain payload mass.

CALLBACKS

Input values trigger
filters on the shared
DataFrame.

Output(..., “figure”) routes
each result to the matching
graph ID.

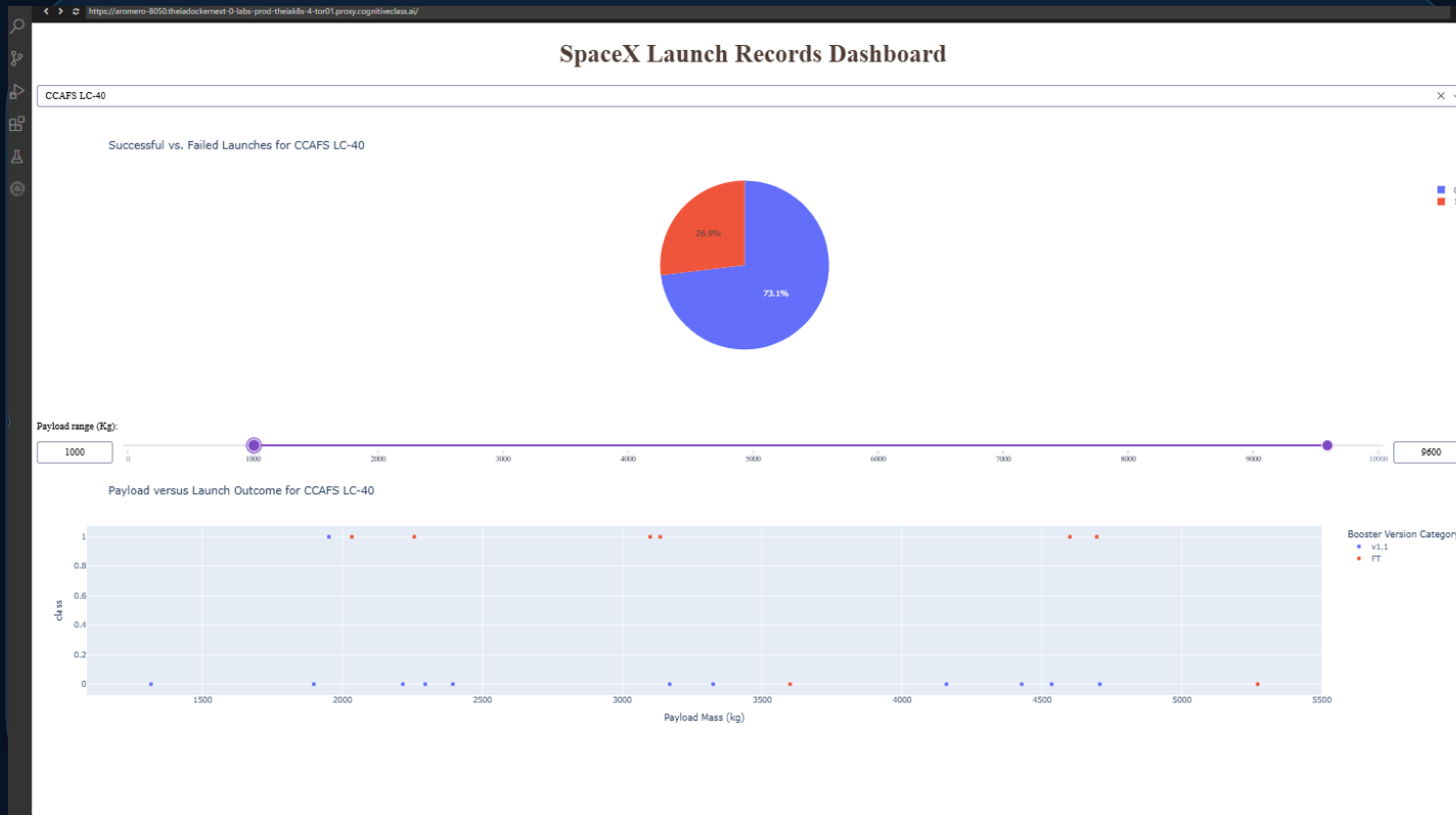
OUTPUTS

Success pie chart

Payload scatter plot

Interactive evidence
updates in real time.

Dash results — site and payload filters work together



Observed insights

41.7%

share of all successes at KSC LC-39A

76.9%

KSC LC-39A success rate

The payload slider removes out-of-range missions; the site dropdown updates both visuals through callbacks.

Predictive analysis — fair model comparison

SPLIT

80% train / 20% test
random_state = 2

STANDARDIZE

Fit scaling on X
reassign transformed X

TUNE

GridSearchCV
10-fold CV

EVALUATE

Holdout accuracy
confusion matrix

Models evaluated



Logistic Regression

C · penalty · solver



Support Vector Machine

C · gamma · kernel



Decision Tree

depth · split · features



K-Nearest Neighbors

neighbors · algorithm · p

Model results — validation lead, test tie, clear error mode

Verified accuracy

Logistic Regression

CV 84.6%

Test 83.3%

SVM — sigmoid

CV 84.8%

Test 83.3%

SVM has the strongest validation score; both verified models classify 15 of 18 test records correctly.

Confusion matrix

Actual fail

3

3

3

false positives

Actual land

0

12

0

false negatives

Predicted fail

Predicted land

Main risk: predicting a landing when the stage actually fails.

Conclusions and innovative next steps

What the evidence supports

- KSC LC-39A leads both success volume and site success rate
- Launch outcomes improve across the flight sequence
- SVM gives the best validation score; Logistic Regression matches its test accuracy
- False positives deserve the most operational attention

Beyond the template

1

COST-AWARE THRESHOLD

Penalize false positives more heavily than false negatives.

2

CALIBRATED PROBABILITY

Report landing probability—not only a binary class.

3

DRIFT MONITORING

Retrain as new booster versions and mission profiles appear.

Recommendation: deploy as decision support, not an autonomous go/no-go rule.